



ScienceDirect

Optics & Laser Technology

Date: December 2025

[Show more](#) ✓[Research article](#) [Full text access](#) [Get rights and content](#) ↗[Full length article](#)

Direct welding of glass and aluminum alloy using nanosecond fiber laser with beam oscillation

Jinming Ma ^{a c}, Meng Jiang ^{a b}  , Xi Chen ^{a c}  , Ao Chen ^{a b}, Xuan Su ^c, Yanbin Chen ^{a c}, Peng He ^{a c} [Show more](#) ✓

View PDF

[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.optlastec.2025.114011](#) ↗

Article

[Recommended articles](#)[Cited by \(1\)](#)[Metrics](#)

Abstract

In this work, glass and aluminum alloy were directly welded using a novel process of nanosecond fiber laser with beam oscillation. The feasibility of achieving reliable joining of glass and aluminum alloy without surface polishing or external pressure was explored via changing the beam oscillation strategy. The results showed that a defect-free glass/aluminum alloy joint could be fabricated with a 40μm lap gap using optimized parameters: a laser power of 25W, a scanning speed of 50mm/s, an oscillation frequency of 750Hz, and an oscillation amplitude of 150μm. The welded joints exhibited a maximum shear strength of 10.95MPa, which is 6 times higher than that without beam

Outline

[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Materials and experimental procedure](#)[3. Results and discussion](#)[4. Conclusion](#)[CRediT authorship contribution statement](#)

oscillation. Fractographic analysis results showed that beam oscillation homogenized laser energy distribution, which could result in an uniform melting and intermixing of glass and aluminum alloy. The enhanced strength of the weld joint was attributed to the improved mechanical interlocking and reduced cracks in the glass modification zone. This work presents a reliable approach for the direct joining of glass and aluminum alloy without the need for surface polishing or external pressure, ensuring high welding efficiency.

Keywords

Nanosecond laser welding; Beam oscillation; Glass; Aluminum alloy; Non-optical contact; Mechanical property

[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

Glass is widely used in aerospace, automobile, biomedical engineering, optical sensing, and other important fields due to their superior optical feature, good mechanical properties, and desirable corrosion resistance [1], [2], [3], [4], [5], [6]]. Currently, glass is often used in conjunction with other materials, such as metals. The most effective use of the composite structure of glass and metal require the production of structurally sound, defect-free and reproducible welds. Several methods, including riveting [7], adhesion [8], soldering [9,10], and diffusion welding [11] were attempted to join glass and metal. These methods, however, often struggle to meet the stringent requirements for composite structures, resulting in joints with inadequate strength or reliability [12]. It is challenging to achieve desirable joining because of the significant difference between the physical and chemical properties of glass and metal.

Declaration of competing interest

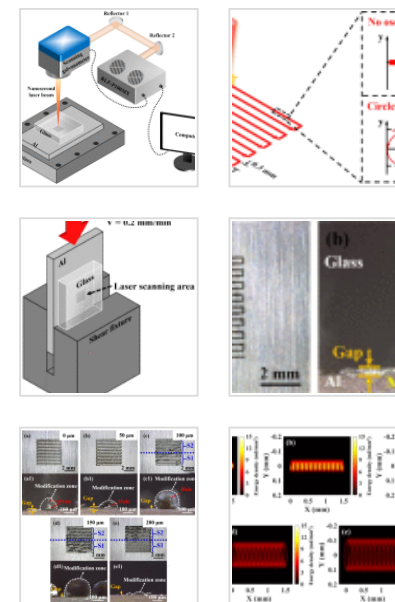
Acknowledgement

Data availability

References

[Show full outline](#) 

Figures (15)



[Show 9 more figures](#) 

Tables (3)

 Table 1. Chemical composition list of sod...

 Table 2. Chemical composition list of 60...

 Table 3. Physical property parameters of...

Ultra-short (femtosecond, picosecond) or short (nanosecond) pulse laser provides a potential pathway to achieve reliable joining of glass and metal due to its tiny heat-affected zone, which can reduce the thermal deformation risk and enhance the joint precision and quality. During the ultra-short pulse laser welding process, a high-density plasma can be formed in materials via multi-photon ionization, resulting in an instant melting and connection [[13], [14], [15], [16], [17]]. Short pulse lasers rely on heat conduction to precisely control energy input and action time for orderly surface melting and connection [[18], [19], [20]]. Among them, nanosecond laser welding is notable as it can balance welding quality, equipment cost, and process complexity, making it possible for large-scale industrial production. Despite its advantages, short pulse laser welding of glass and metal still faces challenges, particularly in maintaining a small welding gap [[21], [22], [23], [24]]. The welding of glass and metal has strict requirements on the contact gap. Excessive gaps can severely compromise welding quality, leading to insufficient connection strength or defects. To overcome this, various methods have been proposed to improve the contact state between glass and metal. Huo et al. achieved optical contact between quartz glass and 304 stainless steel through a series of measures before welding, and used nanosecond lasers to achieve high-quality connections [25,26]. Li et al. proposed a method to stably bond 304 stainless steel and soda lime glass together by leveraging the surface tension of the interstitial fluid, maintaining the two materials within the optical contact range ($<200\text{nm}$) [27]. Then nanosecond laser welding was carried out, and the shear strength of the weld joints reached 9.47MPa. A pre-welding method was subsequently proposed to stabilize optical contact between soda lime glass and 304 stainless steel. Generally, in order to improve the welding success rate, enhance the welding strength and avoid cracks, sample clamps are required to compress the gap between the two materials [[28], [29], [30], [31]]. Nguyen et al. utilized nanosecond laser for the welding of aluminosilicate glass and copper [32]. Before welding, the copper surface underwent a polishing treatment to decrease its surface roughness. A clamping force of 70N was exerted during the welding process. However, under the combined action of the clamping force and the welding thermal stress, crack defects are more likely to occur. Meanwhile, the welding efficiency is also reduced. Therefore, what is currently

needed but not available is an efficient and reliable process for glass and metal welding with low requirements for contact gap and preparation before welding.

During the nanosecond laser welding process, the melting state—referring to the degree of material melting, the uniformity of the molten pool, and the dynamic changes at the solid–liquid interface—plays a critical role in determining the joining quality. In the high-power laser welding fields, beam oscillation has been proved as an effective method to change the laser energy distribution and regulation the molten pool for achieving high-quality joining [[33], [34], [35], [36], [37]]. Zhao et al. studied the energy distribution of circular beam oscillations at different oscillation frequencies [38]. The results showed that the distribution of laser energy became uniform with the increase of oscillation frequency. When the oscillation frequency increases to a certain threshold, the laser energy distribution tends to be stable. It was important to note that at higher oscillation frequencies, the development of spatter became more prominent. Shi et al. found that with the increase of oscillation frequency and amplitude, the peak value of laser input energy distribution gradually decreased [39]. The peak energy position shifted from the center to the sides, resulting in a more uniform energy distribution over the increased distribution. In addition, due to the change of laser energy distribution, the weld morphology also began to change. Hao et al. showed that with the increase of oscillation amplitude and frequency, welding mode from keyhole mode to conduction mode [40]. These findings indicated that the oscillating laser significantly influenced the energy distribution and the formation of the weld. By integrating nanosecond laser technology with beam oscillation, it is anticipated to augment the adaptability of glass and metal welding processes to contact interstices, thereby enabling high efficiency and quality welding of glass and metal.

In this study, glass and metal are directly welded using nanosecond laser without optical contact. Beam oscillation is used to improve the strength and quality of welded joints by changing the molten state of the materials. The effect of beam oscillation on the weld formation is studied by observing the macroscopic morphology and microstructure of welded joints. The shear strength and fracture morphology of welded joints with

different oscillation parameters are discussed. This method offers remarkable convenience and great application potential, providing a new research direction for the welding of glass and metal materials.

2. Materials and experimental procedure

Soda lime glass and 6061 aluminum alloy were selected as the base material for nanosecond laser welding. The 6061 Al exhibited a surface roughness (Ra) of 1.78μm, which is higher than typical values for glass-metal welding [22,41]. The dimensions of 6061 Al and glass were 50mm×25mm×2 mm and 20mm×20mm×2 mm, respectively. There was no polishing treatment before welding, only the surface of the material was cleaned with ethanol, and then wiped with a lint-free lens tissue. The chemical compositions of soda lime glass and 6061 Al are listed in Table 1 and Table 2, respectively. Table 3 summarizes key physical properties, highlighting significant differences in melting points, thermal conductivity, and thermal expansion coefficients between the two materials.

Table 1. Chemical composition list of soda lime glass (wt%).

Component	Fe ₂ O ₃	Al ₂ O ₃	MgO	CaO	Na ₂ O+K ₂ O	SiO ₂
Content	0.1	1	4	13.5	8.1	72

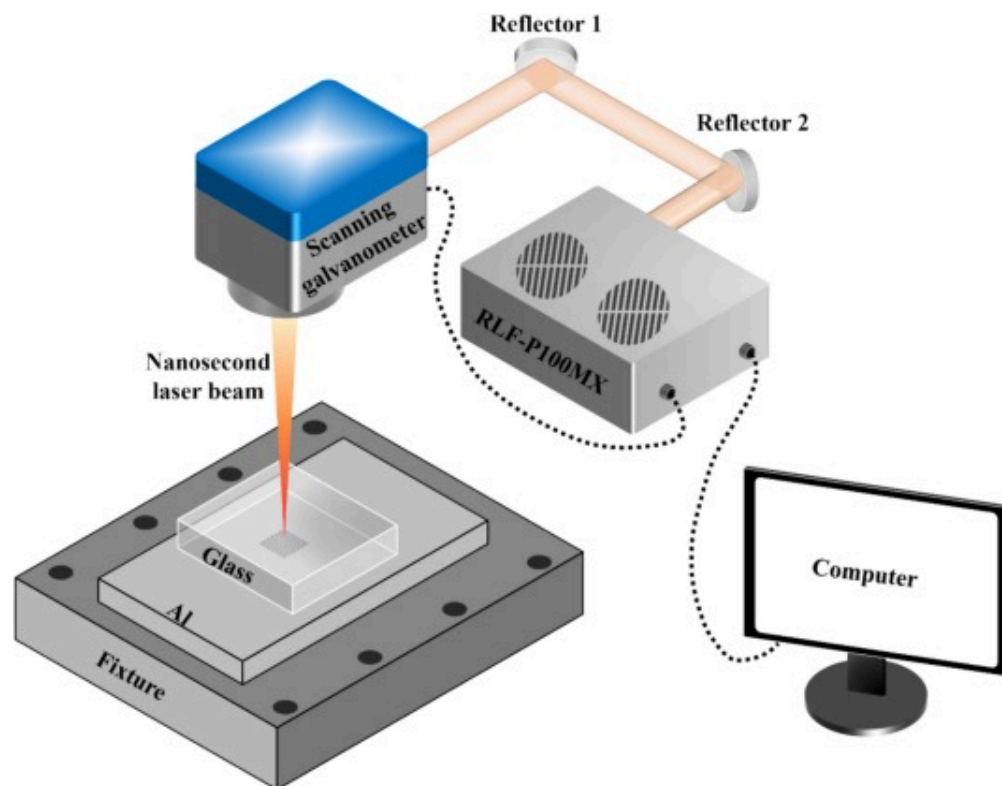
Table 2. Chemical composition list of 6061 Al (wt%).

Element	Cu	Mn	Mg	Zn	Cr	Ti	Si	Fe	Al
Content	0.15–0.4	0.15	0.8–1.2	0.25	0.04–0.35	0.15	0.4–0.8	0.7	Bal.

Table 3. Physical property parameters of experimental materials.

Material	Melting Point (°C)	Density (g·cm⁻³)	Thermal Conductivity (W·m⁻¹·K⁻¹)	Thermal Expansion Coefficient (10⁻⁶·K⁻¹)
Soda lime glass	1425	2.5	0.65	9.5
6061 Al	650	2.75	168	23.5

Fig. 1 depicts the nanosecond laser welding system employed in this study. The system primarily comprises a nanosecond pulsed fiber laser (RLF – P100MX, Wuhan Raycus Fiber Laser Technologies CO., LTD., Wuhan, China), a scanning galvanometer, a field lens, and several reflectors. The output parameters of laser and scanning path of galvanometer are set by computer. The laser emits infrared light with a wavelength of 1064nm, an adjustable pulse duration range from 10 to 350ns, a maximum average power of 100W, and a maximum repetition rate of 2000kHz. The laser beam passes through several reflectors before reaching the scanning galvanometer. The effective focal length of the field lens is 163.5mm. The laser beam is focused into a spot of light with a diameter of 50μm after passing through the lens. The scanning galvanometer is controlled by servo motor to move in the z direction, and the focal plane position is adjusted accurately. As shown in **Fig. 1**, the glass was placed on top of the metal and remained in a natural stacking state during the welding process, without the use of any clamps. The laser was focused on the metal surface through the glass. The pulse duration, pulse repetition rate, and scanning speed were set to 60ns, 250kHz, and 50mm/s, respectively. The gap between the soda lime glass and the 6061 Al was more than 40μm before welding.



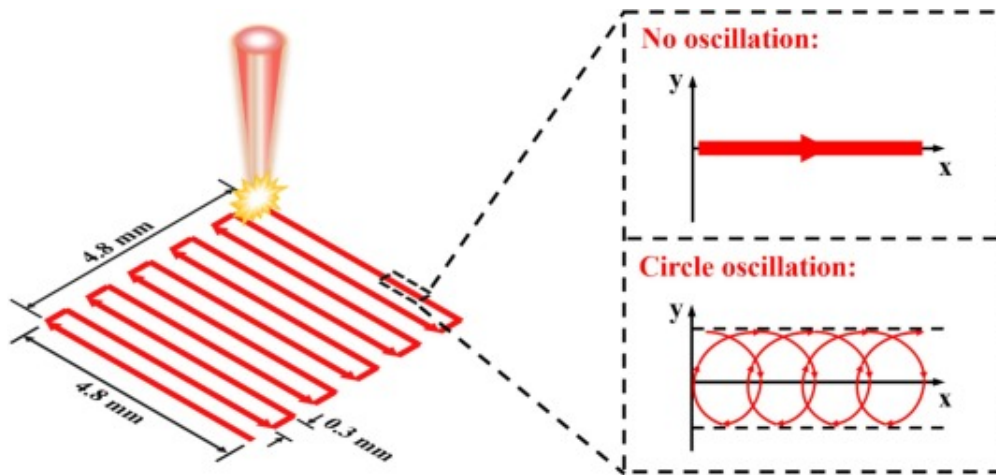
Download: [Download high-res image \(184KB\)](#)

Download: [Download full-size image](#)

Fig. 1. Schematic of nanosecond laser welding of soda lime glass/6061 Al.

Fig. 2 illustrates the beam oscillation path schematic. S-type reciprocating scanning was used for welding, the size of the welding area was $4.8\text{mm} \times 4.8\text{mm}$, and the weld spacing was 0.3mm . Computer-controlled software regulated three parameters: oscillation mode, frequency, and amplitude. Two scanning paths of non-oscillation and circular oscillation were applied to welded samples via software settings. In circular oscillation mode, the laser beam advances along the scanning direction while performing circumferential motion around its travel path center. Key circular oscillation parameters include frequency (oscillations per second) and amplitude (circular diameter). In the enlarged

view of Fig. 2, the x-axis denotes the welding direction, and the y-axis represents oscillation amplitude.



[Download: Download high-res image \(189KB\)](#)

[Download: Download full-size image](#)

Fig. 2. Schematic of oscillating scan path.

To analyze beam oscillation's influence on laser energy distribution during welding, energy distributions under different oscillation parameters were calculated. Assuming the laser travels along the x-axis at constant velocity v , the laser center position in the xy-plane at time t is determined by Equation (1).

$$\begin{cases} x_{\text{center}} = x_{\text{travel}} + x_{\text{oscillation}} = vt + r\cos(2[\text{GSLPI}]ft) \\ y_{\text{center}} = y_{\text{oscillation}} = r\sin(2[\text{GSLPI}]ft) \end{cases} \quad \# \quad (1)$$

among them, x_{center} and y_{center} represent the positions of the laser center on the x-axis and y-axis respectively. x_{travel} is the forward displacement of the laser in the x-axis direction. $x_{\text{oscillation}}$ and $y_{\text{oscillation}}$ are the oscillating displacements of the laser relative to the center of the forward path in the x-axis and y-axis directions at time t respectively. r is the radius of the circular oscillation, and f is the oscillation frequency. Furthermore, laser energy distribution during glass-metal welding is typically approximated by a two-

dimensional Gaussian distribution, as described in Equation (2).

$$E(x, y) = P \times \exp\left(-\frac{(x-x_{\text{center}})^2 + (y-y_{\text{center}})^2}{2\sigma^2}\right) \quad (2)$$

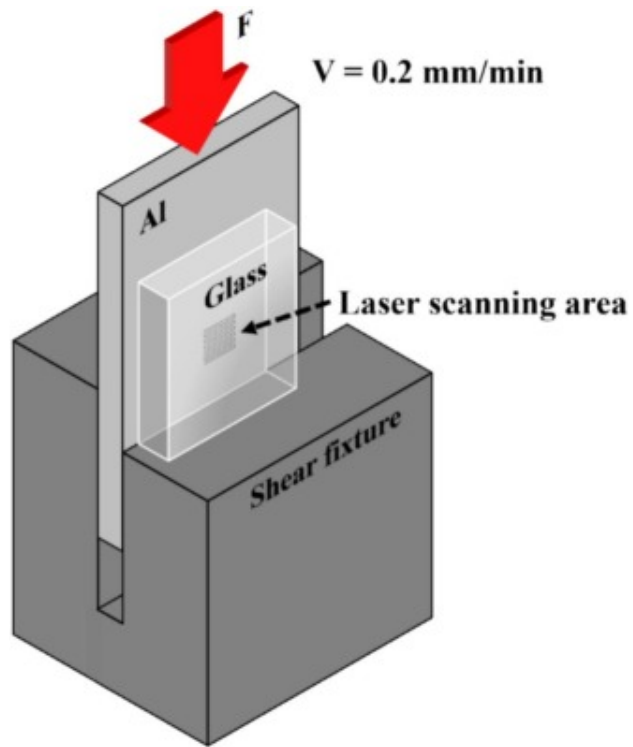
among them, $E(x, y)$ represents the energy density at the coordinates (x, y) . P stands for the laser power. σ is the standard deviation of the Gaussian distribution, which is related to the spot diameter D . Generally, the σ is given by the equation $\sigma = \frac{D}{4}$, and in this work the σ is 12.5.

After the welding was completed, the surface morphology of the welded joints was analyzed with an optical microscope (OM, STEMI 508, ZEISS, Oberkochen, Germany). Metallographic microscope (MM, BX53MRF-S, OLYMPUS, Shinjuku City, Japan) was used to analyze the cross section morphology of welded joints. The microstructure and phase composition of the welded joints were analyzed using a field emission scanning electron microscope (SEM, SU5000, Hitachi, Chiyoda City, Japan) equipped with energy dispersion spectroscopy (EDS).

Shear strength of the welded joints was measured using a universal testing machine (UTM, Instron 5967, Instron, Norwood, MA, USA) at a displacement rate of 0.2 mm/min. Fig. 3 illustrates the shear test schematic. Lap joint shear specimens were employed, with each sample placed in a fixture gap to allow free movement. Frictional effects from the fixture were neglected during testing. The shear sample was slowly displaced over the aluminum alloy, and shear strength was calculated using Equation (3).

$$\tau = \frac{F}{S} \quad (3)$$

where τ is the shear strength (MPa), F is the tested shear force (N), and S is the area of the welding area (23.04 mm² in this study). In the shear strength test, at least three samples were tested for each set of parameters to determine their average strength. After shear strength test, the morphology of soda lime glass side and aluminum side of welded joint fracture was analyzed by laser confocal microscope (LCM, VK-9710, KEYENCE, Osaka, Japan).



[Download: Download high-res image \(43KB\)](#)

[Download: Download full-size image](#)

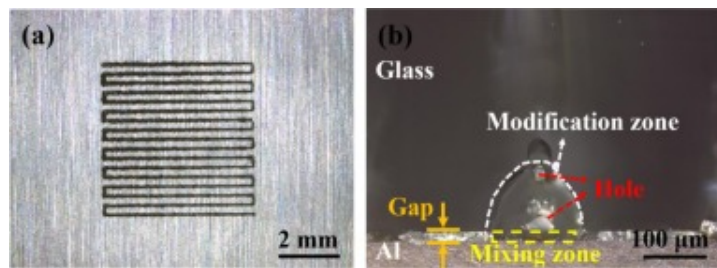
Fig. 3. The test method for shear strength.

3. Results and discussion

3.1. Macroscopic morphology of soda lime glass/Al welded joints

[Fig. 4](#) presents the surface morphology and cross-sectional profile of a nanosecond laser-welded soda lime glass/6061 Al joint. As shown in [Fig. 4 \(a\)](#), a crack-free joint was achieved using this method, with distinct S-type reciprocating scanning tracks visible in the welding area. Due to heat accumulation, the weld width gradually increased along the scanning path. Cross-sectional analysis ([Fig. 4 \(b\)](#)) revealed four distinct zones: the glass

base material, glass modification, mixing, and aluminum base material. Nearly continuous arc-shaped cracks were observed at the interface between the glass base material and modification zone. These cracks arose because molten glass partially collapses during solidification after filling the gap of the glass and metal. The glass modification zone, embedded within the glass substrate, provided mechanical interlocking during shear testing [18,28]. However, glass material used for gap filling led to material deficiency in the modification zone, allowing air to enter and form pores that evolved into larger holes or cracks at the zone apex. Notably, solidification shrinkage of molten materials filled gaps along the laser path and reduced non-scanned gaps to $\sim 15\mu\text{m}$.



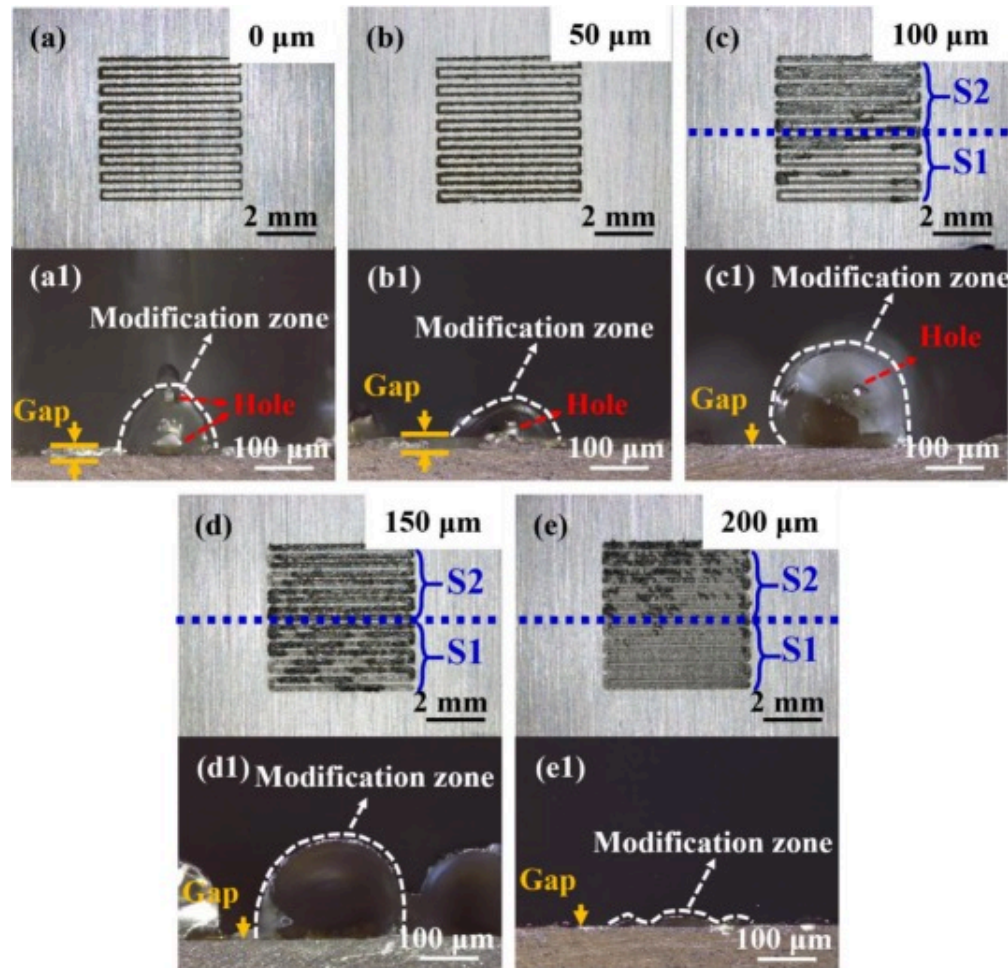
Download: [Download high-res image \(133KB\)](#)

Download: [Download full-size image](#)

Fig. 4. Morphology of soda lime glass/6061 Al joints obtained with nanosecond laser welding: (a) Macroscopic; (b) Cross sectional.

Fig. 5 systematically characterizes the influence of oscillation amplitude on weld morphology evolution. Under a fixed frequency of 750Hz, distinct morphological transitions were observed with increasing amplitudes. According to Fig. 5 (b), when the oscillation amplitude was $50\mu\text{m}$, compared with the case without beam oscillation, there was no significant change in the macro-morphology of the welded joint. The S-type reciprocating scanning path was clearly visible. However, in Fig. 5 (b1), the glass modification zone was reduced. A large number of holes appeared at the bottom of the glass modification zone. Since the oscillation amplitude was equal to the laser spot

diameter of $50\mu\text{m}$, the width of the actual scanning area of the laser did not increase significantly. Due to the solidification contraction effect of the molten glass and metal, the melting of the glass and metal decreased, and the gap increased slightly.



[Download: Download high-res image \(324KB\)](#)

[Download: Download full-size image](#)

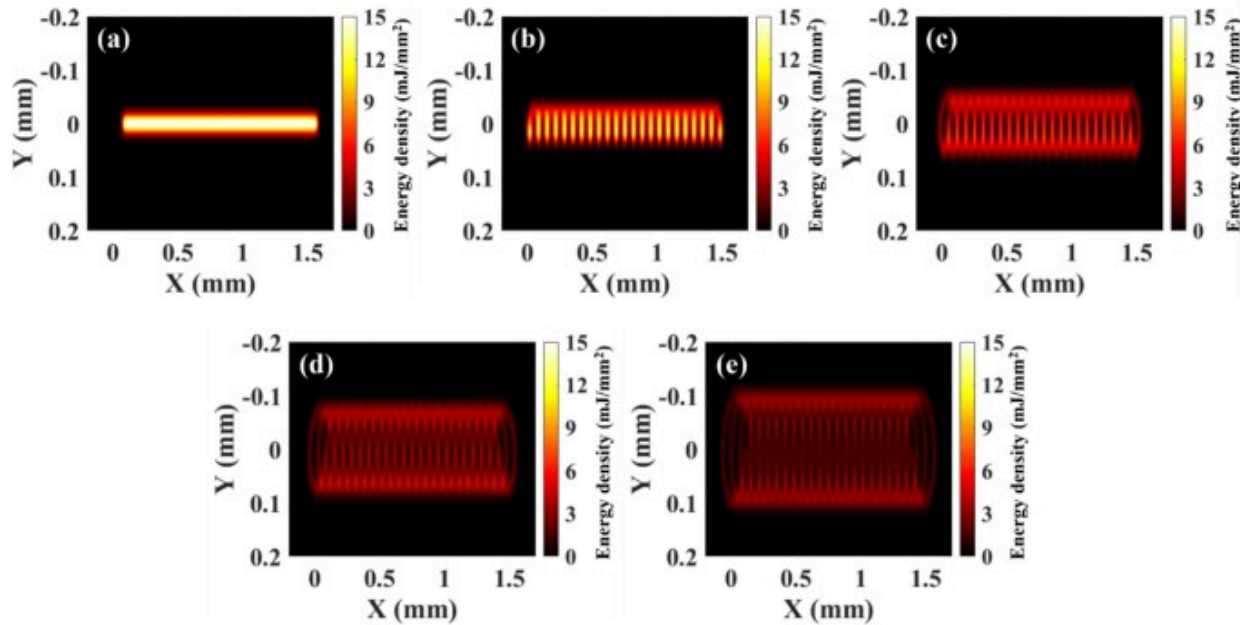
Fig. 5. Macroscopic and Cross section morphology of welded joints with different oscillation amplitude under fixed 750Hz oscillation frequency: (a) $0\mu\text{m}$; (b) $50\mu\text{m}$; (c) $100\mu\text{m}$; (d) $150\mu\text{m}$; (e) $200\mu\text{m}$.

Fig. 5 (c) shows the macro-morphology when the oscillation amplitude increased to 100 μm . The first half (S1) and the second half (S2) of the S-type reciprocating scanning path began to show significant differences. During welding, due to heat accumulation, two welding lines started to contact each other in the local welding area. As can be seen from the cross-section in Fig. 5 (c1), with the increase in the oscillation amplitude, the scanning path became significantly wider, leading to a significant increase in the size of the glass modification area. Nevertheless, there were still holes in the glass modification zone. In addition, the melting of the glass and metal increased significantly, and the gap narrowed noticeably to a few microns because of the solidification and contraction of the molten glass and metal.

As shown in Fig. 5 (d), as the oscillation amplitude continued to climb to 150 μm , the spacing of the scanning path contracted further. In Fig. 5 (d1), the glass modification zone was notably free from any obvious cracks or holes. Fig. 5 (e) shows that when the oscillation amplitude reached 200 μm , the weld spacing of the S-type reciprocating scan path was nearly eliminated. From the cross-section in Fig. 5 (e1), it can be clearly seen that at this juncture, the glass modification zone commenced a rapid decline.

In view of the above phenomenon, Fig. 6 comparatively analyzes the laser energy distribution under varying oscillation amplitudes during the welding process. It can be seen that without beam oscillation, the laser energy distribution was more concentrated, and the energy density was significantly higher compared to when there was beam oscillation. Studies have shown that [42,43], with the decrease of energy density, the welding morphology transitions from mechanical interlocking dominated by nonlinear absorption to metal-dominated heat conduction behaviors (including spattering, wetting, and adhesion). When the overall welding speed was maintained at a constant 50mm/s, as the oscillation amplitude increased from 0 μm to 50 μm , the actual moving speed of the laser spot became faster, and the laser energy density started to decrease, as depicted in Fig. 6 (a) and (b). Therefore, this led to a smaller-sized glass modification zone, and the holes were unable to reach the top, as shown in Fig. 5 (b1). And the laser energy concentration was significantly reduced again in Fig. 6 (c) and (d). Glass could melt more

evenly. A larger glass modification area was obtained. When the laser energy density was further reduced, the laser energy in Fig. 6 (e) was concentrated in the edge and central region of the scanning path, which could correspond to the morphology of the glass modified region in Fig. 5 (e).



Download: [Download high-res image \(241KB\)](#)

Download: [Download full-size image](#)

Fig. 6. Laser energy distribution of welded joints with different oscillation amplitude under fixed 750Hz oscillation frequency: (a) 0 μ m; (b) 50 μ m; (c) 100 μ m; (d) 150 μ m; (e) 200 μ m.

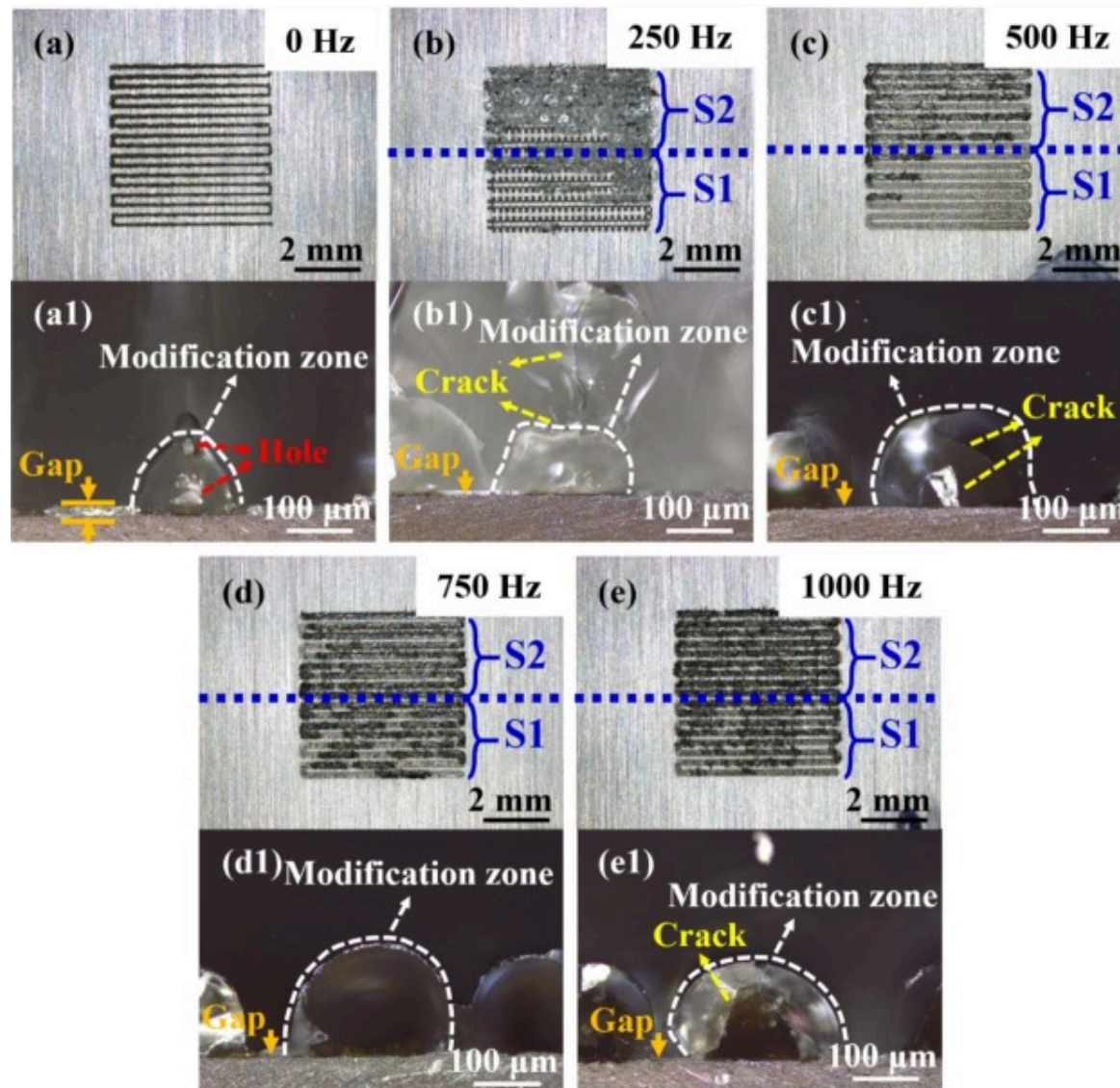
It is worth noting that when the oscillation amplitude increased to over 150 μ m, two welding areas with markedly different colors, namely black and gray weld areas, appeared on the scanning path. By observing the weld cross-sections of different weld areas, it was found that the cross-section of the black weld area typically had a larger glass modification zone, such as Fig. 5 (d1). In contrast, the cross-section of the gray weld

area usually had a smaller glass modification zone, similar to that presented in Fig. 5 (e1). When the oscillation amplitude exceeded 150 μm , the width of the scanning path became larger. This led to the production of more aluminum alloy melt. The greater amount of aluminum alloy melt could form a larger glass modification zone. When it mixed with the molten glass, it resulted in the black color of the weld surface. Due to the welding heat accumulation effect, the black weld area was more commonly distributed in the S2 section of the scanning path, while the gray weld area was more frequently found in the S1 section.

This phenomenon was mainly attributed to the low laser energy density when the oscillation amplitude was large, as shown in Fig. 6. Meanwhile, under the condition of a large contact gap, unstable welding was likely to occur in the S1 section of the scanning path. Particularly when the oscillation amplitude reached 200 μm , the laser energy distribution became more dispersed, the welding heat accumulation effect was significantly weakened, and the difference in the weld appearance between the S1 and S2 sections of the scanning path became more pronounced. Moreover, the proportion of the gray weld area was larger. These findings will be further analyzed in the subsequent research on the fracture morphology of the welded joints.

Fig. 7 illustrated the influence of oscillation frequency on the surface morphology and weld cross section of the welded joint under fixed 150 μm oscillation amplitude. As shown in Fig. 7 (b), when the oscillation frequency was 250Hz, relatively obvious cracks began to appear on the surface of the welded joint. Moreover, due to the cumulative effect of welding heat, cracks were more commonly distributed in the S2 section of the scanning path. This observation can also be verified by the cross-section in Fig. 7 (b1). The cracks originated from inside the glass modification zone and gradually extended upward to the upper surface of the glass. Under this parameter setting, no obvious holes were detected, and the gap between the glass and the aluminum alloy could be narrowed to a few microns. When the oscillation frequency was further increased to 500Hz, as depicted in Fig. 7 (c), the cracks disappeared rapidly. The scanning path became clearer. The black and gray weld areas of the scanning path started to exhibit a distribution pattern similar

to that in Fig. 5 (e), with the black weld areas being more concentrated in S2. By examining the cross section in Fig. 7 (c1), it was found that cracks caused by insufficient melting still persisted inside the glass modification zone.



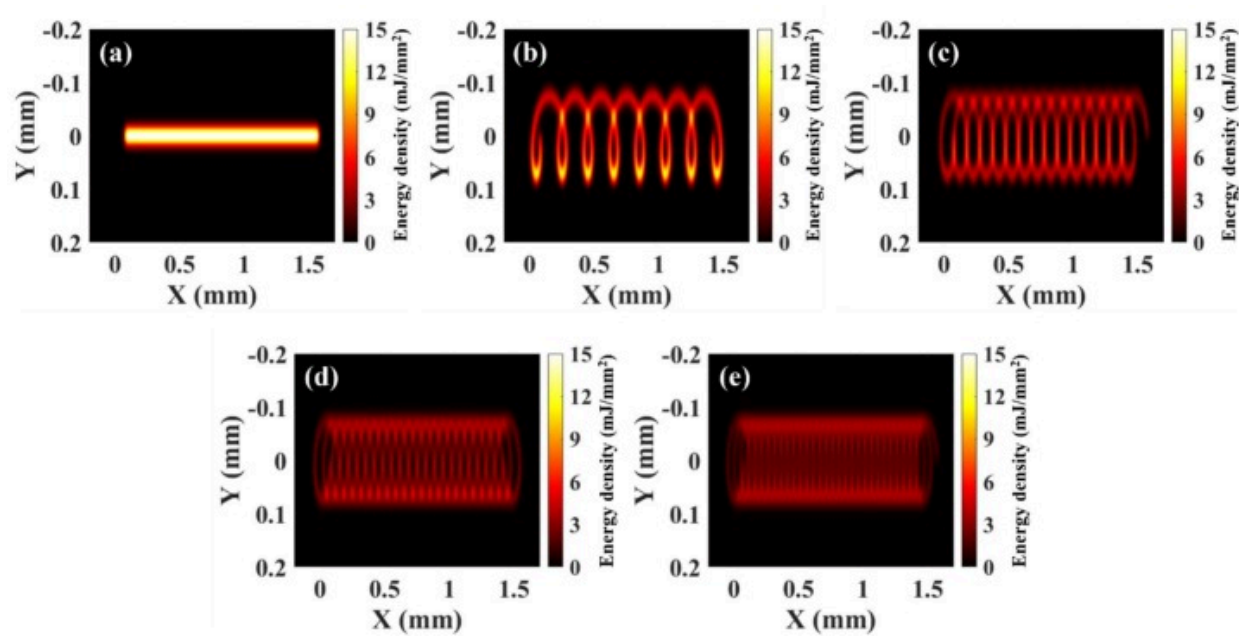
[Download: Download high-res image \(474KB\)](#)

[Download: Download full-size image](#)

Fig. 7. Macroscopic and Cross section morphology of welded joints with different oscillation frequency under fixed 150 μ m oscillation amplitude: (a) 0Hz; (b) 250Hz; (c) 500Hz; (d) 750Hz; (e) 1000Hz.

When the oscillation frequency increased to 750Hz, the cumulative effect of welding heat increased significantly, and the black weld areas of the scanning path expanded notably, as shown in Fig. 7 (d). Comparing Fig. 7 (b1) with (d1), it was evident that as the oscillation frequency increased, the cracks inside the glass modification zone gradually vanished.

According to Fig. 7 (e), when the oscillation frequency continued to rise to 1000Hz, the scanning path was almost entirely composed of black weld areas. This phenomenon indicates a more extensive intermixing of the aluminum alloy melt with the glass. However, from Fig. 7 (e1), it can be seen that when the oscillation frequency was 1000Hz, approximately arc-shaped cracks reappeared inside the glass modification zone. This phenomenon is explained in Fig. 8.



Download: [Download high-res image \(141KB\)](#)

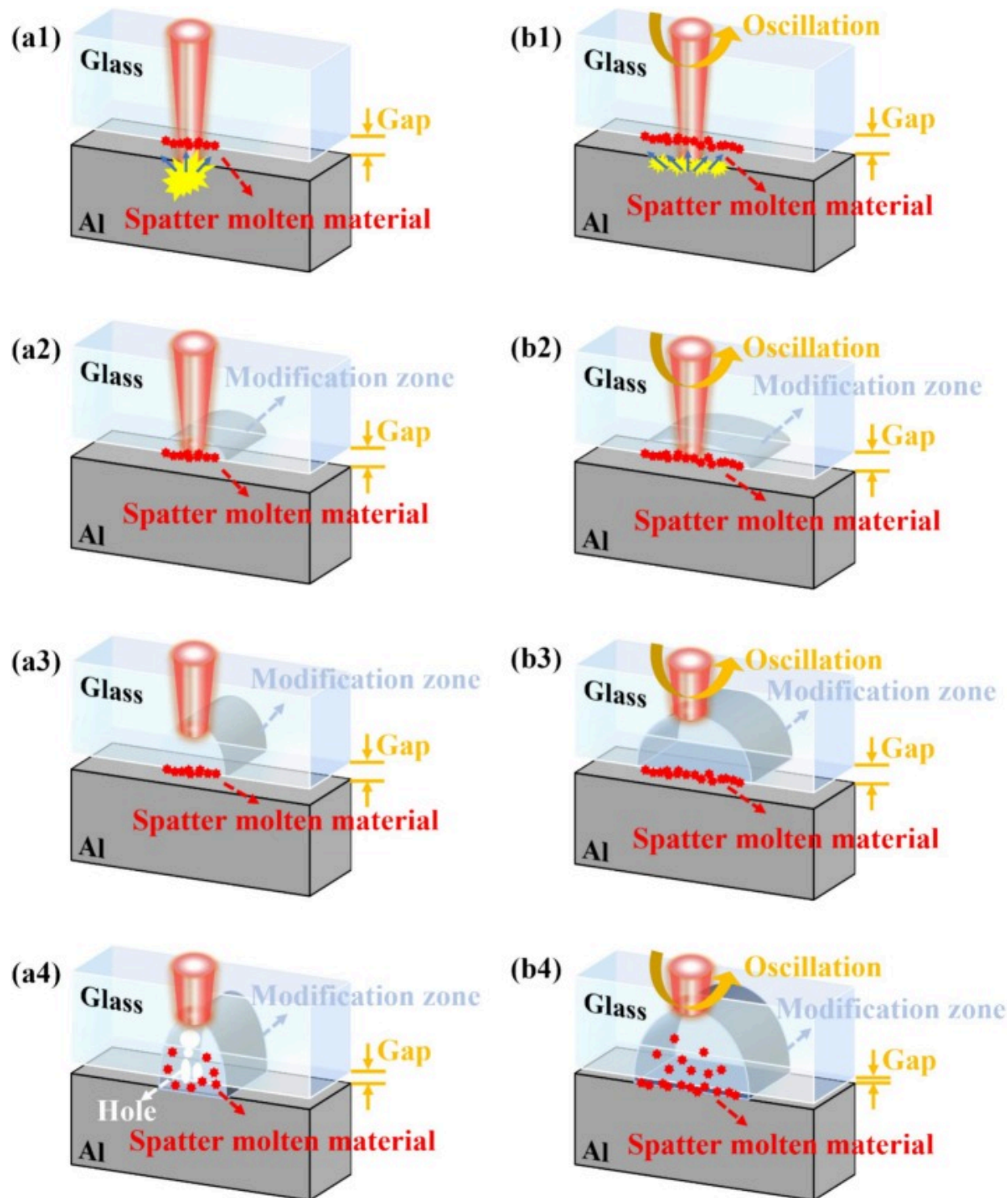
Download: [Download full-size image](#)

Fig. 8. Laser energy distribution of welded joints with different oscillation frequency under fixed $150\mu\text{m}$ oscillation amplitude: (a) 0Hz; (b) 250Hz; (c) 500Hz; (d) 750Hz; (e) 1000Hz.

Fig. 8 systematically quantifies the laser energy distribution at different oscillation frequencies during the welding process. At an oscillation frequency of 250Hz, the laser energy density decreases compared to the non-oscillation condition, but remains concentrated at the corners and intersections of the scanning path, as shown in Fig. 8 (b). This uneven energy distribution leads to non-uniform glass melting, resulting in crack formation. As the oscillation frequency increases, the laser energy distribution becomes more uniform, and the peak energy density decreases, as illustrated in Fig. 8 (b)~(d). This corresponds to the weld morphology in Fig. 7, where cracks gradually disappear. When the oscillation frequency further increases to 1000Hz, the laser energy begins to

concentrate at the edges of the scanning path (Fig. 8 (e)). This causes excessive heat input at the weld edges, leading to the reappearance of cracks.

As discussed earlier, the mechanism underlying nanosecond laser welding of metal and glass involves the melting and resolidification of interfacial materials. The primary absorption mechanism of interfacial materials for nanosecond lasers is collision ionization induced by linear laser absorption. Based on this absorption mechanism, the weld formation process of soda lime glass and 6061 Al nanosecond laser welded joint is shown in Fig. 9.



[Download: Download high-res image \(322KB\)](#)

[Download: Download full-size image](#)

Fig. 9. Schematic diagram of nanosecond laser welding of glass and aluminum alloy: (a1-a4) non-oscillating beam; (b1-b4) oscillating beam.

Fig. 9 (a) illustrates the non-oscillated nanosecond laser welding process between soda lime glass and 6061 Al. During the initial phase (**Fig. 9** (a1)), laser irradiation through the glass substrate induced localized melting of the aluminum surface, with splashing of molten material observed under pulsed laser impact forces. These metallic droplets deposited on the glass underside, concurrently generating sufficient thermal energy to partially melt the glass lower surface. The combined deposition of metallic splatter and modified glass created localized laser-absorbing regions (**Fig. 9** (a2)), initiating progressive formation of the glass modification zone. Subsequent laser irradiation (**Fig. 9** (a3)) triggered rapid expansion of this zone through photothermal conversion, while Marangoni convection and recoil pressure promoted intermixing of glass and aluminum melts. Under gravity-driven flow (**Fig. 9** (a4)), downward migration of the glass modification zone enabled aluminum melt infiltration, establishing interfacial bonding. Concurrently, volumetric shrinkage from melt solidification generated internal voids that coalesced near the zone apex, ultimately forming holes with partial gap reduction through solidification contraction.

Fig. 9 (b) demonstrates the oscillated nanosecond laser welding process between soda-lime glass and 6061 aluminum alloy. During the initial phase (**Fig. 9** (b1)), laser irradiation through the glass substrate melted the aluminum surface, mirroring the non-oscillated process. However, beam oscillation significantly expanded the deposition range of aluminum melt on the glass underside (**Fig. 9** (b2) and (b3)), generating an enlarged glass modification zone. This expanded glass modification zone enhanced gap reduction through volumetric shrinkage during solidification, as evidenced by comparative cross-sectional analyses in **Fig. 5**, where increased oscillation amplitude progressively enlarged

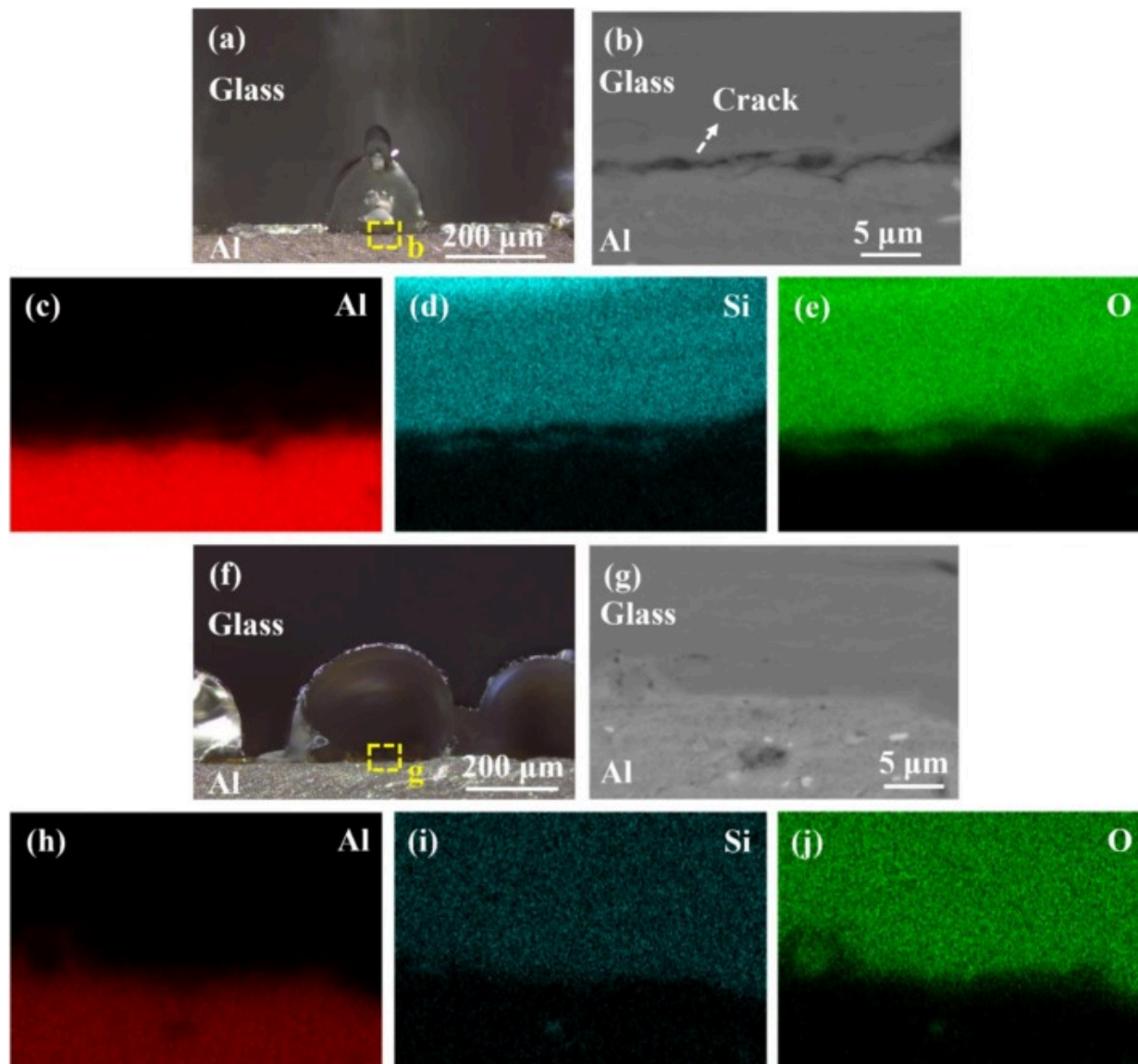
the glass modification zone while suppressing internal holes, achieving complete hole elimination at 150 μ m amplitude. Oscillation frequency critically influenced glass modification zone integrity: lower frequencies (<750Hz) caused incomplete melting with vertically propagating cracks, whereas excessive frequencies reinitiated cracking despite intermediate frequency ranges (750Hz) achieving crack-free interfaces, as shown in Fig. 7. These observations confirm optimal weld morphology occurs at 150 μ m amplitude with 750Hz oscillation frequency.

To deeply analyze the effect mechanism of beam oscillation on the weld quality, the discussion of interface formation will be expanded below. Joint formation between the metal and glass is initiated by localized metal melting induced by transmitted laser energy; metal droplets produced under pulsed or high-intensity conditions impinge on the glass rear surface and deposit, creating locally absorbent regions that promote subsequent heating and modification/melting of the glass. Continued laser energy input expands the modified region via photothermal conversion. Within the molten pool, strong Marangoni convection and recoil-pressure driven flows enhance mixing between the molten metal and the molten/modified glass, promoting the development of a mixed zone and mechanical interlocking at the interface. During cooling and solidification, volumetric shrinkage of the molten material can produce voids or pores at the top of the modified layer; the tendency for pore formation depends critically on the melt's ability to fill gaps and on conditions for gas escape. Beam oscillation alters the instantaneous energy distribution at the interface: an appropriate oscillation amplitude and frequency can homogenize the delivered energy, encouraging more uniform metal deposition and deeper penetration into the modified region, thereby enlarging the modified volume and suppressing longitudinal cracking and large top-surface pores. Conversely, excessive energy dispersion or local edge concentration caused by unsuitable oscillation parameters may promote melt discontinuity or reintroduce cracking and porosity. In summary, joint strength is governed by two coupled contributions: the intrinsic chemical/ metallurgical bonding across the interface and the mechanical interlocking

between the modified zone and the substrates. The above analysis results are consistent with those in references [42,43].

3.2. Microstructure, of soda lime glass/Al welded joints

To investigate beam oscillation's microstructural influence, SEM analysis compared mixing zones in welded joints under non-oscillated and circular-oscillation conditions, as shown in Fig. 10. Magnified cross-sections (Fig. 10 (b) and (g)) demonstrate contrasting interfacial characteristics: stable glass-aluminum bonding was achieved without oscillation, yet accompanied by interfacial micro-cracks, whereas beam oscillation eliminated these defects through enhanced interfacial integrity. The mechanism is attributed to energy distribution differences. Without beam oscillation, concentrated laser energy preferentially absorbed at the glass modification zone's upper region induced rapid vertical growth, concurrently causing insufficient melting at the zone's bottom. In contrast, circular oscillation promoted uniform energy deposition, enabling complete aluminum melt penetration and crack suppression. EDS surface mapping confirms elemental interdiffusion at the bonding interface under both conditions.



Download: [Download high-res image \(318KB\)](#)

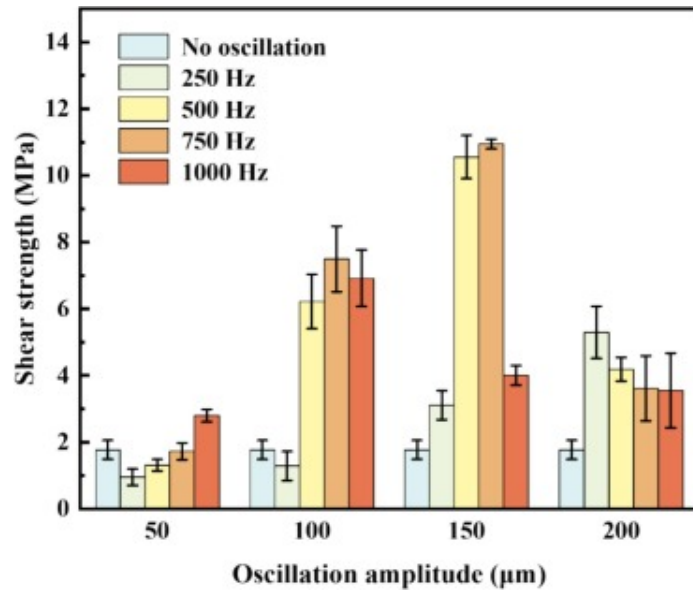
Download: [Download full-size image](#)

Fig. 10. Microstructure of soda lime glass/6061 Al welded joints: (a) Cross sectional morphology of welded joints without beam oscillation; (b) Enlarged view of the mixing zone in (a); (c–e) EDS surface scan of (b); (f) Cross sectional morphology of welded joints

with beam oscillation; (g) Enlarged view of the mixing zone in (f); (h–j) EDS surface scan of (g);

3.3. Mechanical properties of soda lime glass/Al welded joints

Shear testing of soda-lime glass/6061 Al nanosecond laser-welded joints revealed a baseline strength of 1.7MPa without beam oscillation in Fig. 11, aligning with reported values in similar researches [12,44]. Beam oscillation parameters significantly enhanced joint performance, achieving peak shear strength (10.95MPa) at 150 μ m amplitude and 750Hz frequency. At 50 μ m oscillation amplitude, parameter variations exerted negligible effects on joint shear strength due to insufficient displacement magnitude. Shear strength progressively increased with rising oscillation frequency, whereas non-oscillated joints exhibited inferior performance below 750Hz. This stemmed from incomplete melting in the glass modification zone at lower frequencies, promoting crack nucleation and propagation (Fig. 7). At 100 μ m oscillation amplitude, joint strength remained inferior to non-oscillated specimens exclusively at 250Hz. Frequencies exceeding 250Hz induced significant strengthening characterized by initial growth and subsequent minor attenuation. This pattern amplified at 150 μ m amplitude, correlating with progressive volumetric expansion of the glass modification zone as amplitude increased from 100 μ m to 150 μ m. The attenuation mechanism stemmed from laser energy concentration at scanning path edges under elevated frequencies, promoting crack nucleation in the modification zone. These cracks degraded mechanical interlock strength between modified and base material zones, as substantiated by shear fracture morphology analysis.



[Download: Download high-res image \(112KB\)](#)

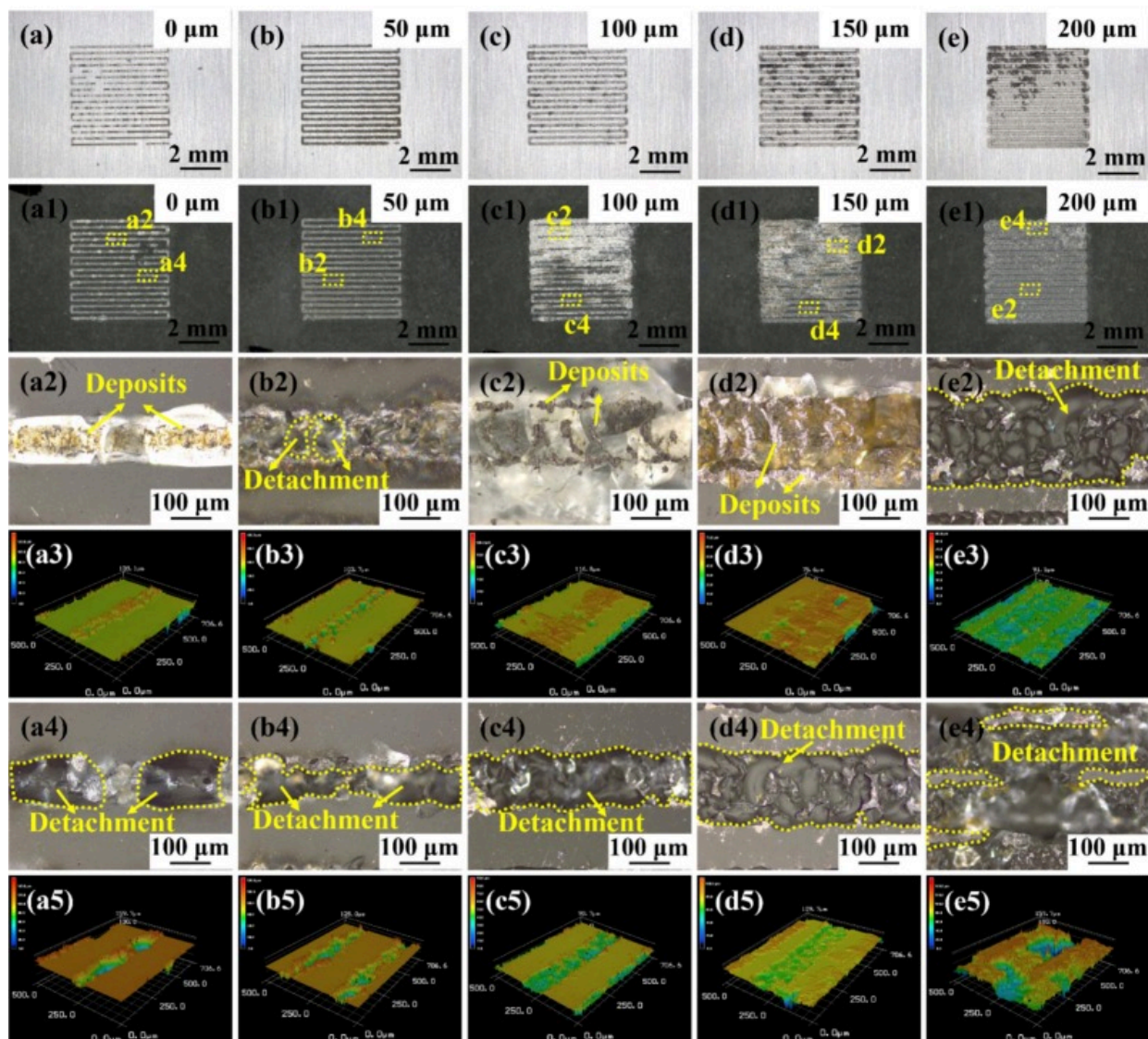
[Download: Download full-size image](#)

Fig. 11. Shear strength of soda lime glass/6061 Al welded joints with different oscillation parameters.

Notably, when the oscillation amplitude was 150μm, the shear strength of welded joints at all tested oscillation frequencies exceeded that of non-oscillating joints. Conversely, when the amplitude increased to 200μm, joint shear strength generally decreased. A gradual decline in shear strength was observed with increasing oscillation frequency. This behavior is primarily attributed to reduced glass melting caused by more dispersed laser energy at the 200μm amplitude. [Section 3.1](#) confirms that while higher frequencies enhanced energy homogeneity, their synergy with 200μm amplitude critically compromised heat concentration, thereby weakening interfacial bonding.

To elucidate beam oscillation effects on nanosecond laser-welded soda lime glass/6061 Al joints, fracture morphology analysis was conducted shear testing. [Fig. 12](#) demonstrates fracture characteristics at 750Hz with varying oscillation amplitudes. Non-oscillated

specimens exhibited discontinuous bonding features in the scanning paths. This instability originated from pre-weld interfacial gaps causing unstable aluminum melt deposition, manifested as localized glass-side discontinuities. Beam oscillation implementation eliminated these defects, achieving homogeneous glass-aluminum fusion. Consequently, oscillated joints demonstrated systematically enhanced shear strength compared to non-oscillated counterparts.



[Download: Download high-res image \(676KB\)](#)

[Download: Download full-size image](#)

Fig. 12. Fracture morphology of soda lime glass/6061 Al welded joints with different oscillation amplitude when the oscillation frequency is 750Hz: (a-e) the fracture surface morphology of the Al side; (a1-e1) the fracture surface morphology of the glass side;

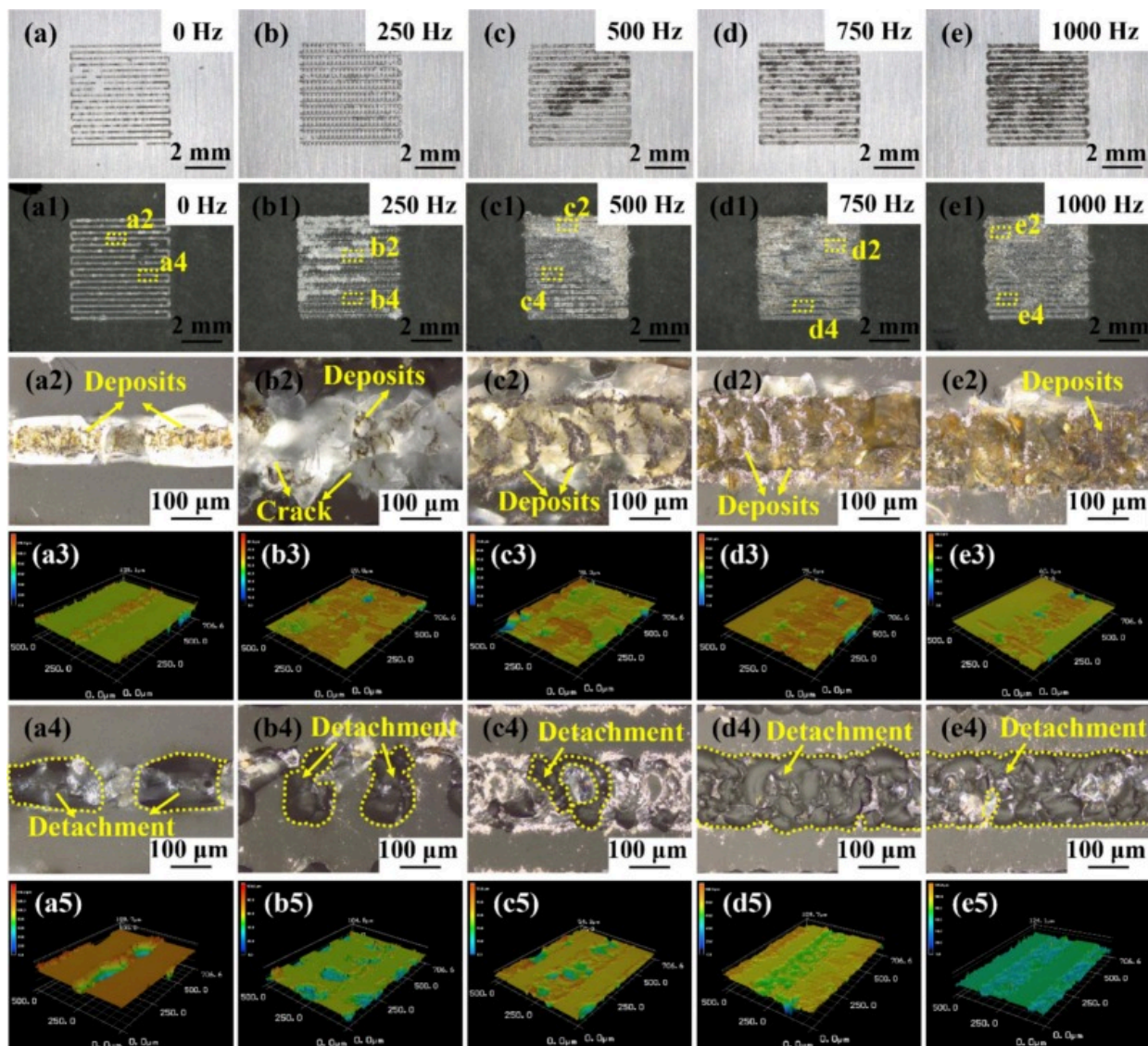
side; (a2-e2) the partial enlargement of (a1-e1); (a3-e3) the 3D morphology of (a2-e2); (a4-e4) the partial enlargement of (a1-e1); (a5-e5) the 3D morphology of (a4-e4);

At different oscillations amplitude of 750Hz, the local area of the fracture surface on the glass side is amplified. Analysis revealed multi-mode fracture characteristics in all parameter sets, as evidenced by corresponding 3D morphologies in [Fig. 12](#). The section analysis above shows that joint shear strength derives from dual mechanisms: mechanical interlocking and interface bonding strength. Mechanical interlocking specifically refers to the physical connection between the glass modification zone and the base material, while interface bonding refers to the atomic bonding of the mixing zone. Superior mechanical interlocking, where glass modification zone/base material interface strength exceeds interfacial bonding capacity, induces interfacial fracture ([Fig. 12 \(a2\)](#)). Interfacial fracture surfaces revealed aluminum alloy melt on both the surface and within the glass modification zone, providing evidence for the welding mechanism analyzed in [Section 3.1](#). Conversely, insufficient interlocking triggered bulk glass fracture ([Fig. 12 \(a4\)](#)). Fracture analysis revealed characteristic craters on glass surfaces, validating this failure mechanism.

At an oscillation amplitude of 50 μ m, no interfacial fracture occurred. Fractures propagated entirely through the glass. The reduced glass modification zone volume caused fracture surfaces to migrate closer to the interface ([Fig. 12 \(b2\)](#) and ([b4](#))). When the amplitude was increased to 100 μ m, the interface fracture appeared again ([Fig. 12 \(c2\)](#)). As the glass modification zone expanded, the mixing zone width increased, and glass surface deposits became more prominent. This configuration yielded significantly higher shear strength than non-oscillating joints. At 150 μ m, extensive aluminum alloy melt deposition on the glass lower surface deepened the glass modification zone color to brown ([Fig. 12 \(d2\)](#)). This is attributed to enhanced intermixing of molten aluminum and glass. At 200 μ m, nearly all fractures occurred on the glass side. Notably, cumulative welding heat expanded the glass modification zone, causing large glass fragments to detach along the scanning path ([Fig. 12 \(e4\)](#)). This led to a significant strength reduction.

These results indicate that during shear testing, interfacial fractures withstood much higher loads than glass-side fractures.

[Fig. 13](#) presents fracture characteristics of soda lime glass/6061 Al joints welded at 150 μ m oscillation amplitude with varying frequencies, displaying both macroscopic and magnified views. At this amplitude, regardless of oscillation frequency, fracture modes were dominated by interfacial fracture and glass-side fracture. Specifically, interfacial fractures primarily occurred in the second half (S2) of the scanning path, while glass-side fractures concentrated in the first half (S1). This distribution is attributed to cumulative welding heat, which promoted more thorough glass melting in the S2 section of the scanning path. Enhanced melting improved mechanical interlock strength between the glass modification zone and the base material. At 250Hz, internal cracks were observed within the glass modification zone, accompanied by sparse surface deposits. As oscillation frequency increased, more uniform laser energy distribution enabled thorough glass melting, reducing crack formation. Concurrently, surface deposits on the glass became denser, and the glass modification zone color deepened, as shown in [Fig. 13](#) (b2) and (e2). This provides evidence for improving the homogeneity of glass and aluminum alloy melt.



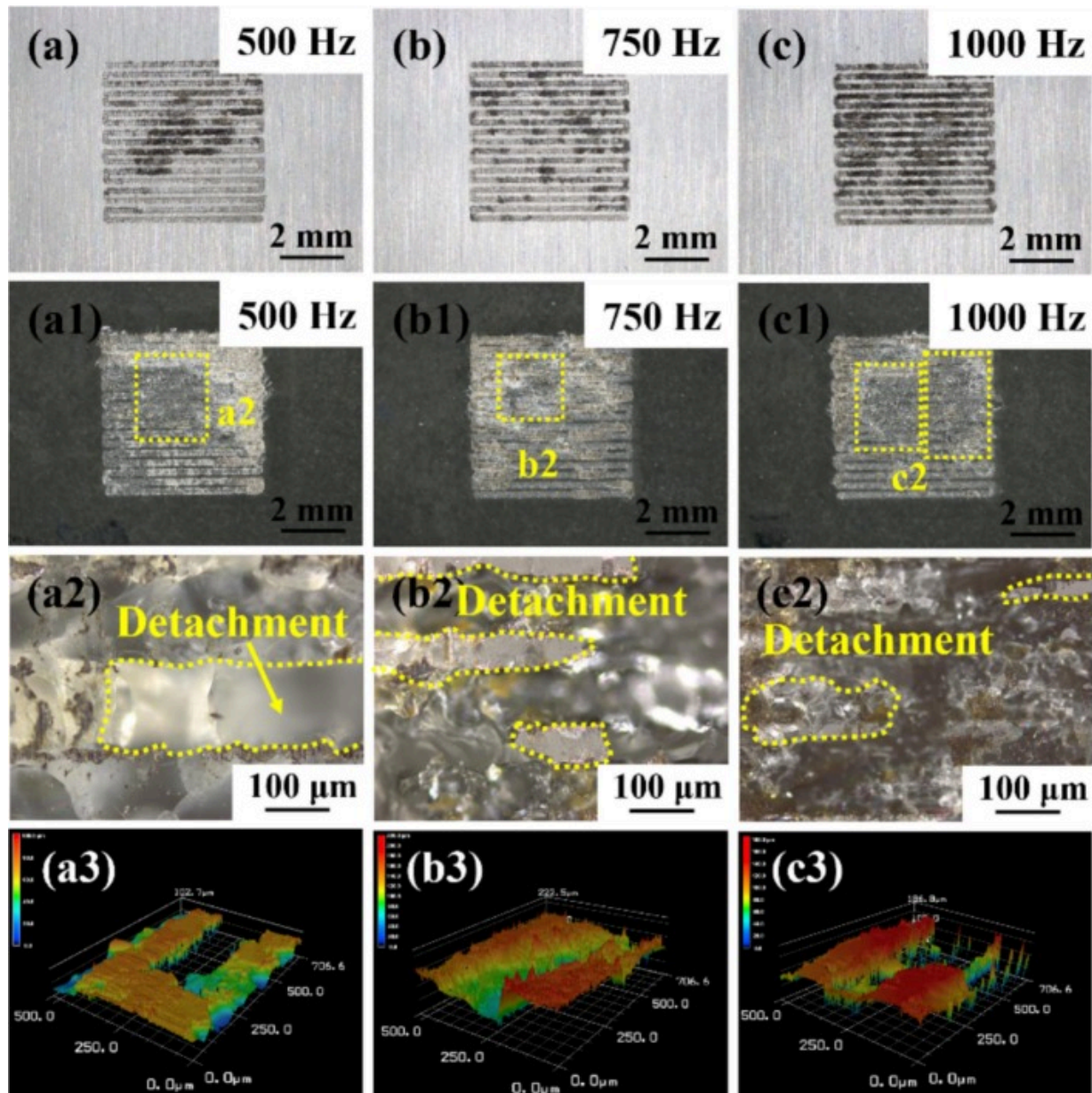
[Download: Download high-res image \(690KB\)](#)

[Download: Download full-size image](#)

Fig. 13. Fracture morphology of soda lime glass/6061 Al welded joints with different oscillation frequency when the oscillation amplitude is 150 μm: (a-e) the fracture surface morphology of the Al side; (a1-e1) the fracture surface morphology of the glass side; (a2-

e2) the partial enlargement of (a1-e1); (a3-e3) the 3D morphology of (a2-e2); (a4-e4) the partial enlargement of (a1-e1); (a5-e5) the 3D morphology of (a4-e4);

Furthermore, at high oscillation frequencies, a third fracture mode emerged: large-scale glass fracture across multiple scanning paths ([Fig. 14](#)). At 500Hz, glass within individual scanning paths began to delaminate locally, as shown in [Fig. 14](#) (a2). However, increasing the frequency to 750Hz caused these loose glass fragments to coalesce into a single large piece spanning multiple paths, which then detached collectively ([Fig. 14](#) (b2)). Cross-sectional analysis ([Fig. 5](#) and [Fig. 7](#)) revealed that large oscillation amplitudes could connect adjacent welds. This phenomenon became more pronounced at 1000Hz, with extensive weld merging observed ([Fig. 14](#) (c2)). Consequently, increasing the oscillation frequency to 1000Hz at a 150 μ m amplitude significantly reduced joint shear strength. This result reaffirms that interfacial fractures sustain much higher shear loads than glass-side fractures during testing.

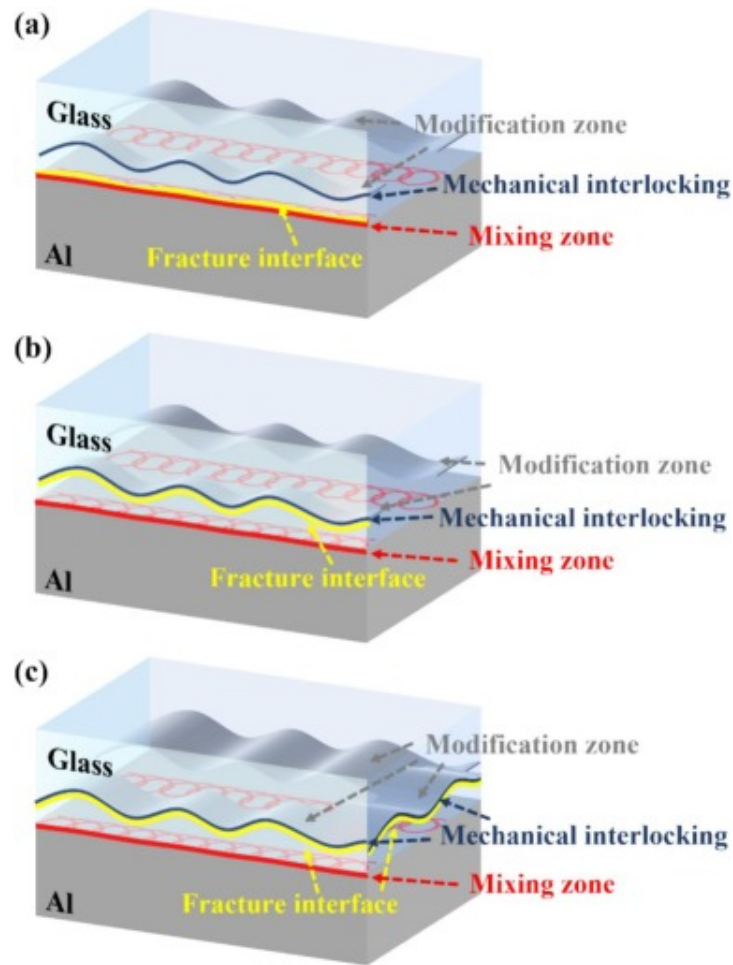


[Download: Download high-res image \(481KB\)](#)

[Download: Download full-size image](#)

Fig. 14. When the oscillation amplitude is 150 μ m and the oscillation frequency is high, the large piece of glass falls off: (a-c) the fracture surface morphology of the Al side; (a1-c1) the fracture surface morphology of the glass side; side; (a2-c2) the partial enlargement of (a1-c1); (a3-c3) the 3D morphology of (a2-c2); (a4-c4) the partial enlargement of (a1-c1); (a5-c5) the 3D morphology of (a4-c4);

The shear strength of welded joints is governed by interfacial strength and mechanical interlocking between the glass modification zone and base metal, producing distinct fracture modes. Three primary fracture modes were observed: interfacial fracture, glass-side fracture, and large-scale glass fracture across multiple scanning paths. Interfacial fracture occurs when mechanical interlock strength between the glass modification zone and base material exceeds interfacial strength (Fig. 15(a)). Conversely, glass-side fracture prevails when interfacial strength dominates (Fig. 15(b)). Large-scale glass fracture across multiple paths arises exclusively at high oscillation amplitudes and frequencies, where adjacent glass modification zones merge (Fig. 15(c)). During shear testing, interfacial fractures withstood the highest shear loads, whereas large-scale glass fractures across multiple paths exhibited the lowest load-bearing capacity. To optimize joint performance in practical applications, interfacial fracture should be prioritized by optimizing laser energy distribution.



[Download: Download high-res image \(115KB\)](#)

[Download: Download full-size image](#)

Fig. 15. Schematic of different fracture modes.

The shear strength of glass-metal welded joints is influenced by multiple factors. Beyond the parameters studied here, pre-welding contact state and surface pretreatment significantly influence joint performance. In previous studies, Ji et al. and Zhang et al. did not grind or polish samples prior to welding [12,45]. Instead, they cleaned material surfaces with alcohol and maintained a natural glass-metal superposition without

clamping pressure during welding, which was a methodology consistent with this work. In contrast, Jia et al. achieved shear strengths up to 6.5MPa by applying clamping force and performing multiple picosecond laser scans on non-optically contacted soda lime glass [46]. To enhance joint strength, Wang et al. polished metal surfaces and utilized clamping force during welding [47]. Li et al. improved pre-welding contact via liquid surface tension, achieving optical contact between glass and metal and producing high strength joints [27].

The shear strength of glass-metal welded joints obtained in this study is significantly higher than that of non-optically contacted joints reported in previous studies. Specifically, this strength is nearly equivalent to joints produced using liquid surface tension assistance and pre-welding treatment. Notably, the beam oscillation welding method proposed here achieves comparable mechanical performance under simple surface cleaning protocols. Beyond clamping force and liquid surface tension strategies, this approach offers a novel, efficient, and economical solution for non-optically contacted glass-metal welding.

4. Conclusion

This study explored the feasibility of direct soda-lime glass/6061 aluminum alloy joining via nanosecond pulsed laser welding with beam oscillation without optical contact. The effect of beam oscillation on the macroscopic morphology, microstructure and mechanical properties was symmetrically investigated. The main conclusions are as follows:

- 1) The nanosecond laser welded joints of glass and aluminum alloy could be macroscopically divided into four zones: glass base material, glass modification, mixing, and aluminum base material. Holes and microcracks were respectively observed in the glass modification zone and in the mixing zone of the weld joint fabricated without beam oscillation. The defects in the weld zone resulted in a shear strength of only 1.7MPa.

- 2) The beam oscillation could significantly reduce the defects in the weld zone of glass and aluminum joints. With a slight increase of oscillation amplitude to 150 μ m, the glass modification zone expanded and the number and volume of the hole defects reduced. When the oscillation amplitude was larger than 150 μ m, the rapidly decreased glass modification zone would deteriorate weld joint. And the oscillation frequency showed a similar trend with oscillation amplitude.
- 3) The optimal oscillation parameters for achieving maximum joint strength were identified as an oscillation amplitude of 150 μ m and an oscillation frequency of 750Hz. Under these conditions, the welded joints exhibited a maximum shear strength of 10.95MPa, which was six times higher than that without beam oscillation.
- 4) Beam oscillation could homogenize laser energy distribution and promote a uniform melting and intermixing of glass and aluminum alloy, which could result in a defect-free glass modification zone with enhanced mechanical interlock with the glass base material. The improved strength of the weld joint was attributed to the enhanced mechanical interlocking and reduced cracks in the glass modification zone and mixing zone.

CRedit authorship contribution statement

Jinming Ma: Writing – review & editing, Writing – original draft, Data curation. **Meng Jiang:** Writing – review & editing, Funding acquisition. **Xi Chen:** Project administration, Funding acquisition. **Ao Chen:** Writing – review & editing. **Xuan Su:** Resources, Methodology. **Yanbin Chen:** Supervision, Project administration. **Peng He:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work is supported by the National Key Research and Development Program of China (Grant No. [2023YFB3407800](#)), the National Natural Science Foundation of China (Grant No. [52205349](#)), the National Fundamental Research Funds (Grant No. [JCKY2023204B003](#)), and the National civil space pre-research project and State Key Laboratory of Precise Welding and Joining, Harbin Institute of Technology ([AWJ-22-T-04](#)).

Data availability

Data will be made available on request.

References

- [1] Z.H. Xiao, S.J. Yu, Y.M. Li, S.C. Ruan, L.B. Kong, Q. Huang, Z.R. Huang, K. Zhou, H.B. Su, Z.J. Yao, W.X. Que, Y. Liu, T.S. Zhang, J. Wang, P. Liu, D.Y. Shen, M. Allix, J. Zhang, D.Y. Tang
Materials development and potential applications of transparent ceramics: a review
Sci. Eng. R-Rep., 139 (2020), p. Mat, [10.1016/j.mser.2019.100518](#) 
[Google Scholar](#) 
- [2] T. Benitez, S.Y. Gomez, A.P.N. de Oliveira, N. Travitzky, D. Hotza
Transparent ceramic and glass-ceramic materials for armor applications
Ceram. Int., 43 (2017), pp. 13031-13046, [10.1016/j.ceramint.2017.07.205](#) 
 [View PDF](#) [View article](#) [View in Scopus](#)  [Google Scholar](#) 
- [3] L.J. Zhou, F. Kato, N. Nakamura, Y. Oshikane, A. Nagakubo, H. Ogi
MEMS hydrogen gas sensor with wireless quartz crystal resonator
Sensor Actuat B-Chem, 334 (2021), [10.1016/j.snb.2021.129651](#) 

[Google Scholar ↗](#)

- [4] W.H. Tuan, S.K. Lee
Eutectic bonding of copper to ceramics for thermal dissipation
Applications-a review
J. Eur. Ceram. Soc., 34 (2014), pp. 4117-4130, [10.1016/j.jeurceramsoc.2014.07.011 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [5] M.H. Ahmed, K. Yoshihara, B. Mercelis, K. Van Landuyt, M. Peumans, B. Van Meerbeek
Quick bonding using a universal adhesive
Clin. Oral Invest., 24 (2020), pp. 2837-2851, [10.1007/s00784-019-03149-8 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] A.A. Siraji, Y. Zhao
High-sensitivity and high-Q-factor glass photonic crystal cavity and its applications as sensors
Opt. Lett., 40 (2015), pp. 1508-1511, [10.1364/OL.40.001508 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [7] F. Lambiase, M. Durante, A.D. Ilio
Fast joining of aluminum sheets with Glass Fiber Reinforced Polymer (GFRP) by mechanical clinching
J. Mater. Process. Technol., 236 (2016), p. 241, [10.1016/j.jmatprotec.2016.04.030 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [8] F. Marchione, G. Chiappini, P. Munafò
Effect of temperature and relative humidity on the shear performance of double-lap adhesive joints between steel and glass adherends
J. Build. Eng., 45 (2022), Article 103546, [10.1016/j.jobbe.2021.103546 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [9] S. Fukumoto, M. Minami, A. Soeda, M. Matsushima, M. Takahashi, Y. Yokoyama, K. Fujimoto

Microstructural development at weld interface between Zr-based glassy alloy and stainless steel by resistance micro welding

J. Phys. Conf. Ser., 379 (2012), Article 012027, [10.1088/1742-6596/379/1/012027](https://doi.org/10.1088/1742-6596/379/1/012027) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [10] Z. Yang, J.W. Stevenson, K.D. Meinhardt
Chemical interactions of barium–calcium–aluminosilicate-based sealing glasses with oxidation resistant alloys

Solid State Ion., 160 (2003), pp. 213-225, [10.1016/S0167-2738\(03\)00160-7](https://doi.org/10.1016/S0167-2738(03)00160-7) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [11] L. Hu, Y. Xue, H. Wang
Glass-Cu joining by anodic bonding and soldering with eutectic Sn-9Zn solder

J. Apply. Compd., 789 (2019), pp. 558-566, [10.1016/j.jallcom.2019.02.257](https://doi.org/10.1016/j.jallcom.2019.02.257) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [12] G. Zhang, G. Cheng
Direct welding of glass and metal by 1 kHz femtosecond laser pulses

Appl. Optics, 54 (2015), pp. 8957-8961, [10.1364/AO.54.008957](https://doi.org/10.1364/AO.54.008957) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [13] H. Chen, J. Duan, Z. Yang, W. Xiong, L. Deng
Picosecond laser seal welding of glasses with a large gap

Opt. Express, 27 (2019), pp. 30297-30307, [10.1364/OE.27.030297](https://doi.org/10.1364/OE.27.030297) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [14] E.H. Penilla, L.F. Devia-Cruz, A.T. Wieg, P. Martinez-Torres, N. Cuando-Espitia, P. Sellappan, Y. Kodera, G. Aguilar, J.E. Garay
Ultrafast laser welding of ceramics

Science, 365 (2019), pp. 803-808, [10.1126/science.aaw6699](https://doi.org/10.1126/science.aaw6699) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [15] Z. Yang, C. Tian, H. Ren, X. Wei, H. Shen
Welding threshold in ultrafast laser welding of quartz glass and 304 stainless steel
Opt. Laser Technol., 181 (2025), Article 111622, [10.1016/j.optlastec.2024.111622](https://doi.org/10.1016/j.optlastec.2024.111622) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [16] X. Yu, Q. Long, Y. Chen, Y. Liu, C. Yang, Q. Jia, C. Wang
Laser transmission welding of dissimilar transparent thermoplastics using different metal particle absorbents
Opt. Laser Technol., 150 (2022), Article 108005, [10.1016/j.optlastec.2022.108005](https://doi.org/10.1016/j.optlastec.2022.108005) ↗
 [View PDF](#) [View article](#) [Google Scholar](#) ↗
- [17] P. Sopeña, A. Wang, A. Mouskeftaras, D. Grojo
Transmission laser welding of similar and dissimilar semiconductor materials
Laser Photonics Rev., 16 (2022), Article 2200208, [10.1002/lpor.202200208](https://doi.org/10.1002/lpor.202200208) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [18] Y. Feng, R. Pan, T. Zhou, Z. Dong, Z. Yan, Y. Wang, P. Chen, S. Chen
Direct joining of quartz glass and copper by nanosecond laser
Ceram. Int., 49 (2023), pp. 36056-36070, [10.1016/j.ceramint.2023.08.285](https://doi.org/10.1016/j.ceramint.2023.08.285) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [19] H. Nguyen, C.K. Lin, P.C. Tung, J.R. Ho
Characterizations of laser transmission welding of glass and copper using nanosecond pulsed laser
Int. J. Adv. Manuf. Tech., 130 (2024), pp. 2755-2770, [10.1007/s00170-023-12838-4](https://doi.org/10.1007/s00170-023-12838-4) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [20] J. Xu, Q. Jiang, J. Yang, J. Cui, Y. Zhao, M. Zheng, J.P. Oliveira, Z. Zeng, R. Pan, S. Chen
A Review on Ultrafast Laser Microwelding of Transparent Materials and Transparent Material–Metals

Metals, 13 (2023), p. 876, [10.3390/met13050876](https://doi.org/10.3390/met13050876) ↗

[Google Scholar](#) ↗

- [21] I. Miyamoto, K. Cvecek, M. Schmidt, Advances of Laser Welding Technology of Glass-Science and Technology-, J. Laser Micro. Nanoen. 15 (2020), <https://doi.org/10.2961/jlmn.2020.02.1001>.

[Google Scholar](#) ↗

- [22] M. Yang, Q. Jiang, X. Zhang, M. Wu, T. Zhang, R. Pan, P. Li, S. Wang, J. Yang
Effects of surface roughness on the microstructure and mechanical properties of dissimilar sapphire/Invar36 alloy joints made by ultrashort pulsed laser micro-welding

J. Laser Appl., 36 (2024), [10.2351/7.0001513](https://doi.org/10.2351/7.0001513) ↗

[Google Scholar](#) ↗

- [23] R. Pan, D. Yang, T. Zhou, Y. Feng, Z. Dong, Z. Yan, P. Li, J. Yang, S. Chen
Micro-welding of sapphire and metal by femtosecond laser
Ceram. Int., 49 (2023), pp. 21384-21392, [10.1016/j.ceramint.2023.03.267](https://doi.org/10.1016/j.ceramint.2023.03.267) ↗



[View PDF](#)

[View article](#)

[View in Scopus](#) ↗

[Google Scholar](#) ↗

- [24] J. Zhan, Y. Gao, J. Sun, W. Zhu, S. Wang, L. Jiang, X. Li
Mechanism and optimization of femtosecond laser welding fused silica and aluminum

Appl. Surf. Sci., 640 (2023), Article 158327, [10.1016/j.apsusc.2023.158327](https://doi.org/10.1016/j.apsusc.2023.158327) ↗



[View PDF](#)

[View article](#)

[View in Scopus](#) ↗

[Google Scholar](#) ↗

- [25] J. Huo, B. Zhang, C. Li, M. Luo, T. Chen, L. Guo, A. Luo, N. Zhao, J. Li, Q. Zhang
The mechanism of the welding between silica glass and 304 stainless steel using nanosecond fibre laser

Sci. Technol. Weld. Join., 28 (2023), pp. 407-414, [10.1080/13621718.2023.2167568](https://doi.org/10.1080/13621718.2023.2167568) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [26] J. Huo, J. Yuan, Q. Chen, M. Luo, J. Lu, J. Xu, C. Li, A. Luo, J. Li, Q. Zhang
Welding reinforcement between silica glass and stainless steel using nanosecond fiber laser with chromium interlayer
Opt. Lasers Eng., 172 (2024), Article 107877, [10.1016/j.optlaseng.2023.107877](https://doi.org/10.1016/j.optlaseng.2023.107877) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [27] C. Li, J. Tan, M. Luo, W. Chen, Y. Huang, J. Gu, N. Zhao, J. Li, H. Yang, Q. Zhang
High shear strength welding of soda lime glass to stainless steel using an infrared nanosecond fiber laser assisted by surface tension
Opt. Lasers Eng., 161 (2023), Article 107329, [10.1016/j.optlaseng.2022.107329](https://doi.org/10.1016/j.optlaseng.2022.107329) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [28] C. Tan, X. Lu, F. Liu, W. Song, G. Guo, Q. Li, Y. Liu, J. Su, X. Song
Realization of joints of aluminosilicate glass and 6061 aluminum alloy via picosecond laser welding without optical contact
Materials, 17 (2024), p. 4299, [10.3390/ma17174299](https://doi.org/10.3390/ma17174299) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [29] A. de Pablos-Martín, T. Höche
Laser welding of glasses using a nanosecond pulsed Nd: YAG laser
Opt. Lasers Eng., 90 (2017), pp. 1-9, [10.1016/j.optlaseng.2016.09.009](https://doi.org/10.1016/j.optlaseng.2016.09.009) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [30] V. Harinarayana, Y.C. Shin
Experimental and numerical investigation on laser transmission welding of stainless steel 304 L and borosilicate glass
Int. J. Adv. Manuf. Technol., 132 (2024), pp. 2661-2675, [10.1007/s00170-024-13527-6](https://doi.org/10.1007/s00170-024-13527-6) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [31] X. Wei, Z. Yang, C. Tian, H. Ren, H. Shen
Measurement of glass position in ultrafast laser welding of glass and metal

Measurement (2024), Article 115161, [10.1016/j.measurement.2024.115161](https://doi.org/10.1016/j.measurement.2024.115161) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [32] H. Nguyen, C.K. Lin, P.C. Tung, J.R. Ho
Microstructures and mechanical properties of copper-to-glass laser
transmission welding employing a nanosecond pulsed laser

Ceram. Int., 50 (2024), pp. 21788-21799, [10.1016/j.ceramint.2024.03.291](https://doi.org/10.1016/j.ceramint.2024.03.291) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [33] S. Sanati, S.F. Nabavi, R. Esmaili, A. Farshidianfar, H. Dalir
A Comprehensive Review of Laser Wobble Welding Processes in Metal
Materials: Processing Parameters and Practical applications

Lasers Manuf. Mater. Process. (2024), pp. 1-37, [10.1007/s40516-024-00245-w](https://doi.org/10.1007/s40516-024-00245-w) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [34] Z. Wang, M. Gao
Numerical simulations of oscillating laser welding: a review

J. Manuf. Process., 119 (2024), pp. 744-757, [10.1016/j.jmapro.2024.04.001](https://doi.org/10.1016/j.jmapro.2024.04.001) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

- [35] S. Kang, J. Shin
Laser beam oscillation welding of aluminum alloy using the spatially
modulated beam by diffractive optical element (DOE)

J. Manuf. Process., 66 (2021), pp. 387-396, [10.1016/j.jmapro.2021.04.029](https://doi.org/10.1016/j.jmapro.2021.04.029) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [36] S. Ma, X. Chen, M. Jiang, T. Han, J. Wang, Y. Yan, Z. Lei, P. He, Y. Chen
Effect of beam oscillating amplitude on forming quality, microstructure,
and mechanical performance of Al-Mg-Sc alloy fabricated by laser-arc
hybrid additive manufacturing

TWS, 205 (2024), Article 112513, [10.1016/j.tws.2024.112513](https://doi.org/10.1016/j.tws.2024.112513) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [37] M. Jiang, B. Li, X. Chen, T. Han, S. Ma, X. Duan, W. Du, Z. Lei, Y. Chen
Enhanced surface finish and reduced porosity of TiC nanoparticles
reinforced 2219 aluminum alloy deposit fabricated via oscillating laser-arc
hybrid additive manufacturing
J. Manuf. Process., 120 (2024), pp. 414-425, [10.1016/j.jmapro.2024.04.071](https://doi.org/10.1016/j.jmapro.2024.04.071) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [38] X. Zhao, J. Chen, W. Zhang, H. Chen
A study on weld morphology and periodic characteristics evolution of
circular oscillating laser beam welding of SUS301L-HT stainless steel
Opt. Laser Technol., 159 (2023), Article 109030, [10.1016/j.optlastec.2022.109030](https://doi.org/10.1016/j.optlastec.2022.109030) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [39] L. Shi, X. Li, L. Jiang, M. Gao
Numerical study of keyhole-induced porosity suppression mechanism in
laser welding with beam oscillation
Sci. Technol. Weld. Join., 26 (2021), pp. 349-355, [10.1080/13621718.2021.1913562](https://doi.org/10.1080/13621718.2021.1913562) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [40] K. Hao, G. Li, M. Gao, X. Zeng
Weld formation mechanism of fiber laser oscillating welding of austenitic
stainless steel
J. Mater. Process. Technol., 225 (2015), pp. 77-83, [10.1016/j.jmatprotec.2015.05.021](https://doi.org/10.1016/j.jmatprotec.2015.05.021) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [41] R.M. Carter, M. Troughton, J. Chen, I. Elder, R.R. Thomson, M.J. Daniel Esser, D. Esser, R.A.
Lamb, D.P. Hand
Towards industrial ultrafast laser microwelding: SiO₂ and BK7 to
aluminum alloy
Appl. Opt., 56 (2017), pp. 4873-4881, [10.1364/AO.56.004873](https://doi.org/10.1364/AO.56.004873) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [42] T. Zhang, M. Wu, Q. Jiang, J. Yang, J.Y. Xu, M. Yang, X. Zhang, R. Pan, P. Li
Understanding the ultrashort laser microwelding process of ceramics to metals by numerical and experimental investigations
J. Laser Appl., 37 (2025), Article 012004, [10.2351/7.0001638](#) ↗
[Google Scholar](#) ↗
- [43] T. Zhang, J. Yang, M. Yang, M. Wu, R. Zhou, J.N. Tian, Y.X. Zhao, C.W. Tan, X.G. Song
Ultrastrong dissimilar glass-ceramic to Fe36Ni alloy joints using ultrafast laser micro-welding
J. Manuf. Process., 150 (2025), pp. 1108-1117, [10.1016/j.jmapro.2025.06.096](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [44] L. Zhang, Z. Zhu, J. Wen, H. Wu, L. Li, X. Ma
The characteristics and dynamics of fused silica-aluminum alloy welding during mJ-level femtosecond laser
Mater. Design., 239 (2024), Article 112790, [10.1016/j.matdes.2024.112790](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [45] C.H. Ji, Y.J. Huang, X. Chen, J.Y. Jiang, Z.J. Guo, Y. Long
Direct microwelding of dissimilar glass and Kovar alloy without optical contact using femtosecond laser pulses
J. Cent. South Univ., 29 (2022), pp. 3422-3435, [10.1007/s11771-022-5091-9](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [46] X. Jia, K. Li, Z. Li, C. Wang, J. Chen, S. Cui
Multi-scan picosecond laser welding of non-optical contact soda lime glass
Opt. Laser Technol., 161 (2023), Article 109164, [10.1016/j.optlastec.2023.109164](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [47] Y. Wang, Y. Li, S. Ao, Z. Luo, D. Zhang

Welding of 304 stainless steel and glass using high-repetition-frequency femtosecond laser

Mater. Res. Express, 8 (2021), Article 106523, [10.1088/2053-1591/ac30b1](https://doi.org/10.1088/2053-1591/ac30b1) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[View Abstract](#)

Recommended articles

Based on reading popularity

Research article Full text access

Mechanistic optimization of wide-gap ultrafast laser quartz glass welding with plasma dynamics

Ning Jiang, ..., Chuanyang Wang

Journal of Materials Processing Technology • Volume 340 • 2025 • Article 118840



[View PDF](#)

Research article Full text access

Welding of glass and single crystal graphite film using a high repetition fs laser

Hongrui Wang, ..., Minglie Hu

Journal of Non-Crystalline Solids • Volume 647 • 2025 • Article 123268



[View PDF](#)

Research article Full text access

Influence of laser powder bed fusion of high-entropy alloy transition layer on the wetting and spreading behaviour of Al alloy on steel substrate surface

Yijian Zeng, ..., Hua Zhang

Journal of Materials Processing Technology • Volume 340 • 2025 • Article 118872



Short communication Full text access

Micro-welding of non-optical contact sapphire/metal using Burst femtosecond laser

Zhou Li, ..., Ji'an Duan

Materials Letters • Volume 377 • 2024 • Article 137540



Research article Full text access

Micro-welding of sapphire and metal by femtosecond laser

Rui Pan, ..., Shujun Chen

Ceramics International • Volume 49, Issue 13 • 2023 • Pages 21384-21392



Research article Full text access

Microcrackless weld formation under unilateral melting in laser transmission welding of sapphire and stainless steel

Jinhui Yuan, ..., Qingmao Zhang

Journal of Manufacturing Processes • Volume 141 • 2025 • Pages 897-905



Cited by (1)

Research article Open access

Numerical modeling and experimental validation of nanosecond laser welding parameters for glass-aluminum bonding

Wang T., ..., Hossain S.

Journal of Materials Research and Technology • Volume 42 • 2026

[Show abstract](#) ✓

Metrics

Citations

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.